UNIT - III

Deadlocks

A deadlock is a situation in which two or more processes are each waiting for a resource held by another, so none of them can proceed. Four conditions must hold at the same time for a deadlock: mutual exclusion, hold and wait, no preemption and circular wait.

Deadlock Avoidance

The Banker's algorithm avoids deadlock by checking whether granting a request keeps the system in a safe state. A safe state is one in which there exists some order in which every process can finish.

Memory Management

Paging divides physical memory into fixed-size blocks called frames and logical memory into blocks of the same size called pages. Paging removes external fragmentation but can cause internal fragmentation.

Segmentation divides a program into variable-sized logical units such as code, stack and data segments.

CPU Scheduling

Round robin scheduling gives each process a fixed time quantum. When the quantum expires the process is moved to the back of the ready queue. Shortest job first minimises average waiting time but can starve long processes.